Training Vectors: SVM QP Formulation

Training Vectors

$x_1 = (1, 1), \quad y_1 = 1$

$x_2 = (2, 2), \quad y_2 = 1$

$x_3 = (1, 4), \quad y_3 = -1$

$x_4 = (0, 4), \quad y_4 = -1$

Test Vectors

$5 \quad 5 \quad 2$

Use C=1

$$\alpha_1 y_1 + \alpha_2 y_2 + \alpha_3 y_3 + \alpha_4 y_4 = 0$$

$$0 \le \alpha_1 \le C$$
$$0 \le \alpha_2 \le C$$
$$0 \le \alpha_3 \le C$$
$$0 \le \alpha_4 \le C,$$

$$\text{Maximize } \mathcal{L}_D(\alpha) = \sum_{i=1} \alpha_i - \frac{1}{2} \sum_{i=1} \sum_{j=1} \alpha_i \alpha_j y_i y_j X_i^T X_j \Rightarrow \text{Minimize } \mathcal{L}_D(\alpha) = \frac{1}{2} \sum_{i=1} \sum_{j=1} \alpha_i \alpha_j y_i y_j X_i^T X_j - \sum_{i=1} \alpha_i$$

X

$\begin{bmatrix} 1 & 2 & 4 & 4 \end{bmatrix}$

$G = XX$

$\begin{bmatrix} x_4 & 24 & 24 & 17 & 00 \end{bmatrix}$

$y_i y_j$

$\begin{bmatrix} y_4 & -1 & -1 & 1 & 1 \end{bmatrix}$

$H = G$

$\begin{bmatrix} y_4 & -17 & -24 & 17 & 00 \end{bmatrix}$

From Previous Lecture

Minimize:

$$\frac{1}{2}[17\alpha_1 + 2\alpha_1\alpha_2 - 5.5\alpha_1\alpha_3 - 6\alpha_1\alpha_4 + 2\alpha_2\alpha_1 + 4\alpha_2 - 11\alpha_2\alpha_3 - 12\alpha_2\alpha_4$$
$$-5.5\alpha_3\alpha_1 - 11\alpha_3\alpha_2 + 32.5\alpha_3 + 36\alpha_3\alpha_4 - 6\alpha_4\alpha_1 - 12\alpha_4\alpha_2 + 36\alpha_4\alpha_3 + 40\alpha_4] - [\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4]$$

$$0 \le \alpha_1 \le C$$
$$0 \le \alpha_2 \le C$$
$$0 \le \alpha_3 \le C$$
$$0 \le \alpha_4 \le C$$

Subject to:

$$\sum_i \alpha_i y_i = 0$$
$$\alpha_1 + \alpha_2 - \alpha_3 - \alpha_4 - 0$$

Quadratic Equation in α with constraints